50.x.x.x

50. [network] . [network] . [network] . [host]

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

prefix length

50.0.0.0 \8 network ID

50.255.255.255 Broadcast IP

Leader

50.0.0.0 /8 ⟶ N/w ID

50.0.0.1 ⟶ first IP

50.0.0.2

50.0.0.3

50.0.0.255

50.0.1.0

50.0.1.1

50.0.1.2

50.0.1.255

50.0.2.0

50.0.255.255

50.1.0.0

50.255.255.254 ⟶ last IP

[50.255.255.255] ⟶ B.IP

host bit

$2^{24} = 16777216 - 2 = 16777214$ valid IP

Insta (firewall) 5000.0.0.0/24

~~50.1~~ ✗

~~50.2~~ ✗

~~50.3~~ ✗

Broadcast IP

HPN — 50.1, 50.2, 50.3 Unicast

50.255.255.255

N. N. H. H

172.172.172.172

172.172.x.x

172. 172. [network] . [network]

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

172.172.0.0 \16 N/w ID

172.172.0.1

172.172.255.254

172.172.255.255 B.IP

$2^{16} = 65,536 - 2 = 65,534$

N. N. N. H

192.192.192.192 ⟶ 255.255.255.0

192.192.x

192. 192. 192. [network]

0 0 0 0 0 0 0 0

1 1 1 1 1 1 1 1

192.192.192.0 \24

192.192.192.1

192.192.192.254

192.192.192.255

$2^{8} = 256 - 2 = 254$

N/C 50.0.0.10

Subnetmask

It is use to mention n/w position in IP

Condition:- put binary as 1 for n/w bit

50.0.0.10

[network] [host]

1 1 1 1 1 1 1 1

255.0.0.0

172.16.0.10

[network] [network] [host] [host]

1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

255.255.0.0

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